🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.EE.3

The Distributive Property

Lesson 6-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary
Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$3(x + 2)$

$3x + 6$

3(4 + 5) = 3×4 + 3×5 = 12 + 15 = 27 — the 3 gets 'distributed' to both the 4 and the 5

Distributive Property

Write the definition:

$3 \times 4 = 12$

3 and 4 are factors

In 3(x + 2), the 3 is the factor outside the parentheses that multiplies each term inside

Factor

Write the definition:

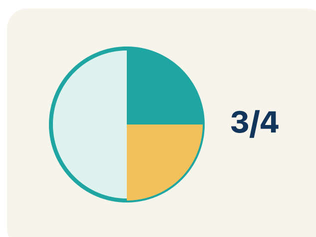
$3(x + 2)$

$3x + 6$

Expand 2(n + 6): multiply 2 × n = 2n and 2 × 6 = 12, so 2(n + 6) = 2n + 12

Expand

Write the definition:



3(x + 4) and 3x + 12 always give the same answer: when x = 2, both = 18; when x = 10, both = 42

Equivalent

Write the definition:

$\underbrace{4}\,x$

coefficient = 4

In 6x + 15, the coefficient of x is 6 — it came from distributing in 3(2x + 5)

Coefficient

Write the definition:

____

____

$\boxed{3x} + \boxed{2x} = \boxed{5x}$

same variable → combine

4x + 2x = 6x (combine coefficients); 4x + 2y cannot be combined (different variables)

Like terms

Write the definition:

____

____

Guided Notes

Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can use the distributive property to expand and factor expressions.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Distributive Property | Factor | Expand | Equivalent | Coefficient | Like terms

👆 *Tap any word to see what it means and a picture.*

1. The rule that $a(b + c) = ab + ac$ is the ____________.

2. To rewrite a sum as a product, like $3x + 12 = 3(x + 4)$, is to ____________.

3. To multiply out a product over a sum, like $3(x + 4) = 3x + 12$, is to ____________.

4. Two expressions that always have the same value are ____________.

5. A number multiplied by a variable is a ____________.

6. Terms with the same variable part are ____________.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Expand $6(n + 3)$ using the distributive property.

A. $6n + 18$

B. $6n + 3$

C. $n + 18$

D. $6n + 9$

Step 1 $6(n + 3) = 6 \times n + 6 \times 3 = 6n + 18$.

✅ **Answer:** A. $6n + 18$

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Expand $4(5 - 2)$ using the distributive property.

A. 12

B. $20 - 2$

C. 9

D. 22

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Expand 3(x + 4) using the distributive property.

A. 3x + 12

B. 3x + 4

C. x + 12

D. 7x

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Light Cue 2: Expand $5(2 + n)$ to set two spotlight banks.

A. $10 + 5n$

B. $10 + n$

C. $7 + 5n$

D. $10 + 5$

Show your work:

2. Light Cue 3: Expand $4(a + 2)$ for the back-row beam.

A. $4a + 8$

B. $4a + 2$

C. $a + 8$

D. $4a + 6$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Use the distributive property to explain why 6 × 98 can be calculated as 6 × 100 − 6 × 2 = 588. Then create your own example of using the distributive property to make multiplication easier.

Sentence starter: 6 × 98 = 6(100 − 2) = 6 × ____ − 6 × ____ = ____ − ____ = ____. My example: ____ × ____ = ____(____ ± ____) = ____.

Show your work:

Reflect — Exit Ticket

Which expression is equivalent to 7(x + 3)?

A. 7x + 21

B. 7x + 3

C. x + 21

D. 7x + 10

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Distributive Property
2. **Notes 2:** Factor
3. **Notes 3:** Expand
4. **Notes 4:** Equivalent
5. **Notes 5:** Coefficient
6. **Notes 6:** Like terms
7. **We Try (notes):** A. 12 *— $4(5 - 2) = 4 \times 5 - 4 \times 2 = 20 - 8 = 12$.*
8. **You Try (notes):** A. 3x + 12 *— $3(x + 4) = 3 \times x + 3 \times 4 = 3x + 12$.*
9. **Try It 1:** A. 10 + 5n *— Distribute the 5 to each term: $5 \times 2 + 5 \times n = 10 + 5n$.*
10. **Try It 2:** A. 4a + 8 *— Distribute the 4 to each term: $4 \times a + 4 \times 2 = 4a + 8$.*
11. **Exit Ticket:** A. 7x + 21 *— $7(x + 3) = 7 \times x + 7 \times 3 = 7x + 21$.*